

Quick Info

- Candidate: [Your Name]
- Institution: [Your University / School]
- Supervisor: [Supervisor Name]
- Period: [Start Date] → [End Date]
- Keywords: Zero Trust, IAM, SIEM, Threat Modeling, Secure SDLC

1. Context & Motivation

Organizations increasingly rely on cloud services, APIs, and distributed workflows. This expands the attack surface and raises the need for consistent identity, access, and monitoring controls. The proposed PFE focuses on designing a practical security blueprint and a prototype implementation aligned with modern best practices.

2. Problem Statement

How can we define and validate a security architecture for a small-to-medium information system that (a) limits lateral movement, (b) detects suspicious activity quickly, and (c) supports secure development and operational practices without excessive cost or complexity?

3. Objectives

- Produce a security requirements baseline (assets, threats, compliance needs).
- Define target architecture (network segmentation, identity, logging, backups).
- Build a minimal prototype demonstrating core controls (e.g., IAM policies, audit logs).
- Evaluate using clear metrics (MTTD/MTTR proxies, coverage of top threats, usability).

4. Scope

In scope: threat modeling, security architecture, IAM, logging/SIEM concepts, secure CI/CD controls, basic incident response playbooks. Out of scope: full enterprise SOC operations, advanced malware analysis, and large-scale production deployment.

5. Methodology

1) Asset inventory and data classification. 2) Threat modeling (STRIDE) and risk ranking. 3) Architecture design (Zero Trust principles). 4) Prototype implementation. 5) Testing (misconfig scenarios, least-privilege checks). 6) Documentation and final defense.

6. Deliverables

- PFE book (state of the art + design + implementation).
- Architecture diagrams and security policies.
- Prototype code/configuration.
- Evaluation report and recommendations.

7. Timeline (Example)

Weeks 1–2: requirements + threat model. Weeks 3–5: architecture + design validation. Weeks 6–8: prototype and security checks. Weeks 9–10: report writing + final review.

Notes / Appendix

This page is reserved for references, diagrams, risk register updates, and supervisor feedback.

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